RIZWAN FAROOQ

Applied Statistician | Climate & Environmental Statistics | Biostatistics & Public Health| Bayesian Modelling | Spatiotemporal and Panel-Data Modelling | Time Series | Computational Statistics|

SUMMARY

An applied Statistician and interdisciplinary researcher specialising in Bayesian, spatial and spatiotemporal modelling, and uncertainty quantification for complex environmental processes and large-scale public-health data with strong computational skills in R, R-INLA, RStan, and Python. Proficient in ArcGIS, SPSS, LaTeX, and Microsoft Office. PhD in Applied Statistics, complemented by 15+ years of experience in applied statistics and real-world data analysis.

PERSONAL INFORMATION

Email	rizwanfarooq@stat.qau.edu.pk; rizwanfarooq04@gmail.com
Google Scholar	https://scholar.google.com/citations?hl=en&user=K3TzuYEAAAAJ
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ResearchGate	https://www.researchgate.net/profile/Rizwan-Farooq-3
Scopus Author ID	https://www.scopus.com/authid/detail.uri?authorId=59559296300
Web of Science Author Record	https://www.webofscience.com/wos/author/record/QAX-1421-2026
Loop Profile	https://loop.frontiersin.org/people/3538619/overview
Date of Birth	September 16, 1989
Address	Bureau of Statistics, 65-A Trade Centre Block, M.A Johar Town, Lahore.
Cell	+92-304-1147646

RESEARCH/PUBLICATIONS/PAPERS/ARTICLES

1. Farooq, R., Niaz, R., Hussain, I. *et al.* Improving drought risk assessment using advanced statistical approaches. *Acta Geophys.* 74, 204 (2026). https://doi.org/10.1007/s11600-026-01918-0

2. Farooq, R., Abd Elwahab, M. E., Niaz, R., Cheng, H., Dalam, M. E. E., Almazah, M. M. A., & Hussain, I. (2026). Monitoring drought occurrence through a count-based statistical modeling. Physics and Chemistry of the Earth, 144. https://doi.org/10.1016/j.pce.2026.104426

3. Farooq, R., Elwahab, M. E. A., Niaz, R., Cheng, H., Dalam, Mhassen. E. E., Almazah, M. M. A., & Hussain, I. (2026). Integrating climatic and topographic drivers in a poisson-GEE framework for modelling drought dynamics. Theoretical and Applied Climatology, 157(6), 353. https://doi.org/10.1007/S00704-026-06235-6

4. Farooq, R., Niaz, R., Hussain, I., ALMAZAH, M. M. A., HILALI, S. O., & CHENG, H. (2025). A Bayesian Framework for Spatiotemporal Analysis of Meteorological Drought Dynamics. Pure and Applied Geophysics. https://doi.org/10.1007/s00024-025-03841-9

5. Farooq, R., Niaz, R., Hussain, I., Almazah, M. M. A., Al-Rezami, A. Y., & Albasheir, N. A. (2025). Bayesian panel modeling of spatiotemporal dynamics in intraseasonal meteorological drought persistence. Stochastic Environmental Research and Risk Assessment. https://doi.org/10.1007/s00477-025-03027-8

6. Sajjad, A., Ahmad, M., Tariq, R., Iqbal, M., Farooq, R., ali, W., & Ali, A. (2025). Assessment of Community Perceptions on Drinking Water Quality and its Implications for Human Health in Islamabad, Pakistan: A Comprehensive Analysis. Desalination and Water Treatment, 101055. https://doi.org/10.1016/j.dwt.2025.101055

7. Farooq, R., Khan, H., Khan, M. A., & Aslam, M. (2020). Socioeconomic and demographic factors determining the underweight prevalence among children under-five in Punjab. BMC Public Health, 20(1). https://doi.org/10.1186/s12889-020-09675-5

Reviews for Journals

- Environmental Earth Sciences
- BMC Psychiatry
- BMC Cancer

WORK EXPERIENCE (Bureau of Statistics Punjab)

Statistical Officer (18-04-2011 to 05-11-2021)| **Assistant Director** (05-11-2021 to-date)

The overall job duties are as follows: -

i) **Lead and supervise statistical workflows** encompassing data collection, data quality control and editing, tabulation, statistical analysis, graphical presentation, interpretation, and technical report preparation, ensuring the production of reliable and analytically useful statistical evidence.
ii) **Manage complex, large-scale statistical datasets and data-production processes**, with responsibility for data quality, completeness, coverage, consistency, timely processing, and dissemination across multiple statistical domains.
iii) **Coordinate large-scale field-based data collection and survey operations**, including monitoring field performance, response rates and coverage, evaluating fieldwork plans, and coordinating with field offices to maintain data quality and representativeness.
iv) **Contribute to the design, implementation and statistical execution of large-scale household, demographic, health and socioeconomic surveys**, providing practical experience in survey methodology, sampling operations, field data quality, statistical processing and evidence generation.
v) **Undertake multidisciplinary statistical work across development-relevant domains**, including agriculture, health, education, industrial production and employment, socioeconomic conditions and population-related statistics, providing extensive experience in analysing heterogeneous real-world datasets.
vi) **Coordinate the preparation and statistical dissemination of major official statistical publications**, including *Punjab Development Statistics, Punjab Health Statistics, Punjab Agriculture Statistics, Punjab Education Statistics,* and the *Monthly Survey of Industrial Production and Employment in Punjab*.
vii) **Collaborate with government departments, field offices and other institutions** on statistical information requirements, data coordination, methodological issues and delivery of official statistical outputs.

AD-HOC SURVEYS

Sr.#	Survey Title	Duration	Designation
1	Post Typhoid Conjugate Vaccine (TCV) Campaign Coverage Survey	19-02-2021 to 05-03-2021	Team Supervisor
2	Multiple Indicator Cluster Survey	01-11-2017 to 10-03-2018	Team Supervisor
3	Punjab Women Economic and Social Well-Being Survey	2017	Member Working Group
4	Home Based Workers Survey	05-11-2016 to 09-12-2016	Team Supervisor
5	Child Labour Baseline Survey	01-08-2016 to 20-10-2016	Team Supervisor
6	Punjab Child Labour Survey (Pre Testing)	31-05-2016 to 04-06-2016	Team Supervisor
7	Food Security Analysis Survey	15-01-2016 to 02-02-2016	Team Supervisor
8	Multiple Indicator Cluster Survey	01-06-2014 to 13-09-2014	Team Supervisor
9	Mother Child Week Survey	28-12-2013 to 03-01-2014	Team Supervisor

TRAININGS

Sr.	Name of Course Attended	Duration	Name of Institute
1	National Capacity-Building Workshop on Statistical Data Analysis of Integrated Disease Surveillance Reference (IDSR) System.	19-02-2024 to 23-02-2024	National Institute of Health (NIH), Islamabad
2	Population Projection	12-11-2018 to 16-11-2018	Population Welfare Tanning Institute, Lahore
3	Cs-Pro Software on Basic Level	24-05-2016 to 26-05-2016	GIZ, Lahore
4	Information Technology Skills	18-04-2016 to 22-04-2016	Management and Professional Development Department (MPDD) Punjab
5	PC-I Preparation	02-09-2013 to 06-09-2013	
6	Financial Management	24-09-2012 to 28-09-2012	
7	Advance Microsoft Office	07-12-2011 to 10-12-2011	

EDUCATIONAL QUALIFICATION

Class/Diploma	Institute/Board/ University	Session	Obtained CGPA/Marks	Total Marks/CGPA	Marks (%)
PhD. (Statistics)	Quaid-i-Azam University, Islamabad	2022-2026	**Title**: Bayesian Spatiotemporal Modelling of Climate Warming and Drought Persistence: A Comprehensive Framework for Prediction and Risk Assessment		
MPhil. (Statistics)	Government College University Lahore	2014-2016	3.13	4.00	78.25%
MSc. (Statistics)	University of Sargodha	2008-2010	3.41	4.00	78.33%
BSc. (Math, Stat, Economics)	University of Sargodha	2006-2008	490	800	61.25%
FSc. (Pre-Engineering)	BISE Sargodha	2004-2006	686	1100	62.36%
Metric (Science)	BISE Sargodha	2002-2004	627	850	73.76%

REFERENCE

1. **Dr. Ijaz Hussain**, Professor (Statistics), Department of Statistics , Quaid-i-Azam University, Islamabad, (Contact Details: +92-51 9064-2183, ijaz@qau.edu.pk)

2. **Dr. Rizwan Niaz**, Assistant Professor (Statistics), Department of Statistics, Kohsar University Murree, Muree (Contact Details: +92-344 5567716, razwanniaz11@gmail.com)
   Currently**:** Postdoctoral research fellow at School of Energy and Environment Science, Yunnan Normal University, Kunming 650500, China.

3. **Dr. Sajid Ali**, Associate Professor (Statistics), Department of Statistics , Quaid-i-Azam University, Islamabad, (Contact Details: +92-51 9064-2185, sajidali@qau.edu.pk)